

SACHIN M V

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OBJECTIVE

Applied AI and Machine Learning Engineer with a Master Degree in Data Science and nearly three years of combined professional experience spanning industrial operations, research analysis and engineering. Skilled in Python, Machine Learning, Computer Vision and Natural Language Processing with hands-on experience developing AI solutions for safety critical applications and multilingual NLP. Seeking opportunities to build intelligent, scalable AI solutions in Machine Learning, Computer Vision, NLP and Industrial AI.

EDUCATION

Jawaharlal Nehru University

Master of Technology in Data Science

New Delhi

2024 – 2026

Mahatma Gandhi University

Bachelor of Technology in Electrical and Electronics Engineering

Kerala

2012 – 2016

TECHNICAL SKILLS

Programming Languages: Python, C, C++, MATLAB, SQL, HTML

Machine Learning & Computer Vision: PyTorch, scikit-learn, OpenCV, YOLOv8, Hugging Face Transformers

Data & Development Tools: NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, Git, Google Colab

Domain Specific Tools: AutoCAD, DIALux, QGIS, MATLAB, Android Studio

Platforms: Linux, Windows, Kaggle, LaTeX, Roboflow

EXPERIENCE

Shift Engineer

Henry and Farad Pvt Ltd — TELK (Transformers & Electricals Kerala Ltd)

Angamaly, Kerala

Aug 2022 – Jul 2024

- Operated a 66 kV substation serving continuous industrial and municipal load, maintaining up time in compliance with IEC/IEEE protection standards and KSEB safety protocols.
- Performed daily load monitoring, switching operations, and condition inspections on transformers, circuit breakers, relays, and differential protection systems.
- Documented fault logs, maintenance records, and incident reports; identified recurring relay mis-coordination patterns and escalated for corrective maintenance.
- Acquired operational understanding of real world grid behaviour that now directly informs research in AI-based fault detection and smart infrastructure monitoring.

Freelance Content Researcher

Reliance Big Synergy — Surya TV

Kochi, Kerala

Dec 2020 – Apr 2021

- Developed factual trivia content for *Anchinodu Inchodinchu* (Malayalam adaptation of *Are You Smarter Than a 5th Grader?*), aired on Surya TV.
- Verified all questions against primary sources including academic texts, archives, and official records to meet broadcast grade accuracy requirements.

Research Analyst

Nexus Consulting — Mazhavi Manorama TV

Aroor, Kerala

Oct 2019 – Mar 2020

- Developed and rigorously fact checked question sets for *Ningalkkum Aakam Kodeeshwaran* (Malayalam adaptation of *Who Wants to Be a Millionaire?*), spanning Science, History, Current Affairs, and Literature across difficulty tiers.
- Managed strict content confidentiality using a proprietary CMS; coordinated on-site with the host and production crew during live studio recordings.

Trainee Electrical Engineer

Tritech Corporation

Kalamassery, Kerala

Jun 2017 – Sep 2017

- Assisted in designing LV wiring layouts, substation schematics, solar connections, and generator installations using AutoCAD and DIALux.
- Conducted load calculations, prepared bill of quantities (BOQ), and participated in site inspections for electrical project commissioning.

ACADEMIC PROJECTS

Rider Assistance Using Smartphones | *Python, scikit-learn, GPS* 2026

- Designed a smartphone-based safety framework for two-wheeler riders using GPS and inertial motion data to estimate collision risk in real time.
- Built a feature engineering pipeline over positional and kinematic signals; trained an ML risk classifier to categorize interaction severity between vehicles.
- Applied spatial filtering to isolate high-risk interactions and generate traffic risk visualizations for dense urban environments.

Malayalam NLP Assistant | *Python, HuggingFace, MarianMT, BERT, Gradio* 2025

- Built a five function NLP tool for Malayalam: bidirectional translation (En ↔ ML via Helsinki-NLP MarianMT), script-to-Manglish transliteration (ITRANS), multilingual sentiment analysis (BERT), and emotion detection (DistilRoBERTa).
- Integrated LangDetect for automatic language identification and pipeline routing; deployed as an interactive Gradio web app.

OLTC Voltage Regulation using Arduino | *Arduino Uno, C, relay modules* 2016

- Simulated On-Load Tap Changer (OLTC) behaviour with AVR control logic and relay switching, achieving $\pm 5\%$ voltage stabilization under variable lab load conditions.

Microcontroller-Based Temperature Controller | *Microcontroller, C, temperature sensor* 2015

- Designed a self regulating heater using a microcontroller temperature feedback loop; programmed adaptive control logic in C for automatic heater intensity adjustment.

CERTIFICATIONS & TRAINING

PDP on Autonomous Vehicle Training Systems — *NITTTR Chennai* (Feb 2026)

AI in Healthcare — *Microsoft* (2026)

Data Analytics Essentials — *Cisco Networking Academy* (2026)

Research Methodology — *UNITAR* (2026)

Electrical System Design and Drafting — *Tritech Design Academy* (2017)

Full certificate list: sachinmavelil.vercel.app

ADDITIONAL

- **Quiz:** Multi-district champion across Science, Social Science, Business, Environment, and Literature; cleared preliminary and mains rounds of competitive examinations.
- **Leadership:** Organized an 8-team intercollegiate charity football tournament as part of a 4-member executive panel who handled registrations, scheduling, venue logistics, budget, and dispute resolution.
- **Collaboration:** Actively participated in the JNU Debate Club, engaging in structured debates and discussions that strengthened critical thinking, public speaking, and evidence-based argumentation.
- **Languages:** English (professional), Malayalam (native), Hindi (working proficiency).
- **Research Interests:** AI for electrical infrastructure and smart grids, ADAS, low resource NLP for Indic languages, and computer vision for industrial inspection.